

■ SurfaceGraphics

`SurfaceGraphics[array]` is a representation of a three-dimensional plot of a surface, with heights of each point on a grid specified by values in *array*.

`SurfaceGraphics[array, shades]` represents a surface, whose parts are shaded according to the array *shades*.

`SurfaceGraphics` can be displayed using `Show`. ■ The same options can be given for `SurfaceGraphics` as for `ListPlot3D`. ■ *array* should be a rectangular array of real numbers, representing *z* values. There will be holes in the surface corresponding to any array elements that are not real numbers. ■ If *array* has dimensions $m \times n$, then *shades* must have dimensions $(m - 1) \times (n - 1)$. ■ The elements of *shades* must be either `GrayLevel[i]` or `RGBColor[r, g, b]`. ■ `SurfaceGraphics` is generated by `Plot3D` and `ListPlot3D`. ■ See page 120. ■ See also: `ListPlot3D`, `Plot3D`, `ContourGraphics`, `DensityGraphics`.